

Arch linux installation

Connect to wifi:

- Verify that with `rfkill` that it is unblocked.
- Run `iwctl` and then

```
device list
stations name scan
station name get-networks
station name connect SSID
exit
```

- Verify that it is connected to wifi with `ping archlinux.org`.
- Problem with `timedatectl`:
 - Type `nano /etc/systemd/timesyncd.conf` and add `NTP=time.google.com`. Then run `systemctl restart systemd-timesyncd`.
 - Verify with `timedatectl`.
- Partitioning
 - Run `lsblk` to see the partition table and then run `cfdisk /dev/name`.
 - Then create three partitions: `swap_partition`, `root_partition` and `boot_partition`; with the types: Linux swap, Linux filesystem, EFI system, respectively. Then select Write and confirm.
 - To format those partitions:

```
mkfs-vfat -F 32 /dev/boot_partition
mkfs.btrfs -f /dev/root_partition
mkswap /dev/swap_partition
```

- Btrfs subvolumes:

```
mount /dev/root_partition /mnt
btrfs su cr /mnt/@
btrfs su cr /mnt/@home
btrfs su cr /mnt/@pkg
btrfs su cr /mnt/@log
btrfs su cr /mnt/@snapshots
umount /mnt
```

- Mountpoints:

```
mkdir /mnt/archinstall
mount -o noatime,compress=zstd:5,discard=async,space_cache=v2,subvol=@ /dev/
root_partition /mnt/archinstall
mkdir -p /mnt/archinstall/home
mkdir -p /mnt/archinstall/var/cache/pacman/pkg
mkdir -p /mnt/archinstall/var/log
mkdir -p /mnt/archinstall/.snapshots
mount -o noatime,compress=zstd:5,discard=async,space_cache=v2,subvol=@home /dev/
root_partition /mnt/archinstall/home
mount -o noatime,compress=zstd:5,discard=async,space_cache=v2,subvol=@pkg /dev/
root_partition /mnt/archinstall/var/cache/pacman/pkg
mount -o noatime,compress=zstd:5,discard=async,space_cache=v2,subvol=@log /dev/
root_partition /mnt/archinstall/var/log
mount -o noatime,compress=zstd:5,discard=async,space_cache=v2,subvol=@snapshots /
dev/root_partition /mnt/archinstall/.snapshots
mkdir -p /mnt/archinstall/boot
mount /dev/boot_partition /mnt/archinstall/boot
swapon /dev/swap_partition
```

- Arch install:
 - Select the mirror, locales, user.
 - Disk configuration: Select Pre-mounted configuration and type /mnt/archinstall.
 - Bootloader: Select grub.
 - Optional repositories: Multilib
 - Additional packages: Select with TAB, then press ENTER.
- Grub configuration:

After installation select chroot, then run:

```
grub-install --target=x86_64-efi --efi-directory=/boot --bootloader-id=GRUB
grub-mkconfig -o /boot/grub/grub.cfg
```

App installation

- Neovim

```
sudo pacman -S neovim npm fzf ripgrep git curl wget npm lazygit yazi
```

- Utils

```
sudo pacman -S brightnessctl playerctl upower power-profiles-daemon wl-clipboard
cliphist unzip foot 7zip udiskie udisks2 gvfs ImageMagick poppler easyeffects
```

- Shell programs

```
sudo pacman -S fish starship zoxide eza github-cli
```

- Cli apps

```
sudo pacman -S btop cava ttyter
```

- Fonts

```
sudo pacman -S ttf-liberation ttf-jetbrains-mono-nerd ttf-jetbrains-mono ttf-font-
awesome noto-fonts noto-fonts-cjk noto-fonts-emoji noto-fonts-extra cantarell-fonts
ttf-nerd-fonts-symbols ttf-nerd-fonts-symbols-common ttf-nerd-fonts-symbols-mono
```

- LaTeX

```
paru -S texlive-full
sudo pacman -S typst zathura zathura-pdf-poppler
sudo pacman -S biber perl-yaml-tiny perl-file-homedir
sudo fmtutil-sys --all
```

- Hyprland

```
sudo pacman -S hyprlock hypridle hyprsunset hyprpicker nautilus tela-circle-icon-
theme-blue swww fuzzel
sudo pacman -S xdg-desktop-portal-gtk xdg-desktop-portal-gnome qt5-wayland qt6-
wayland uwsn
paru -S bibata-cursor-theme grimblast-git matugen-bin
```

- Applications

```
sudo pacman -S firefox spotify-launcher inkscape python-tinycss2 gtksourceview3
```

- Games

```
paru -S protonup-qt
sudo pacman -S lib32-mesa steam
sudo pacman -S gamemode lib32-gamemode
sudo gpasswd -a c5514 gamemode
```

Ricing:

- GTK theme:

```
gsettings set org.gnome.desktop.interface icon-theme "'Tela-circle-blue-dark'"
gsettings set org.gnome.desktop.interface color-scheme 'prefer-dark'
```

- Inkscape window problem:

```
sudo cp /usr/share/applications/org.inkscape.Inkscape.desktop ~/.local/share/
applications/
sudo sed -i 's/^Exec=inkscape/Exec=env GDK_BACKEND=x11 inkscape/' ~/.local/share/
applications/org.inkscape.Inkscape.desktop
update-desktop-database ~/.local/share/applications/
```

Snapshots

- Install snapper dependencies:

```
sudo pacman -S snapper snap-pac grub-btrfs
```

- Generate snapshost subvolume with snapper

```
sudo umount /.snapshots
sudo rm -rf /.snapshots
sudo snapper -c root create-config /
```

- The previous command creates .snapshots but we want to use @snapshots so:

```
sudo btrfs subvolume delete .snapshots
sudo mkdir /.snapshots
sudo mount -a
sudo chmod 750 /.snapshots
sudo chown :wheel /.snapshots
```

- Now we want to create our first snapshot:

```
sudo snapper -c root create -d "***Base system install**"
```

- Now we want to enable some services:

```
sudo systemctl start snapper-timeline.timer
sudo systemctl enable snapper-timeline.timer
sudo systemctl start snapper-cleanup.timer
sudo systemctl enable snapper-cleanup.timer
```

- Enable snapshot grub entries:

- Now enable grub-btrfs service:

```
sudo systemctl enable --now grub-btrfsd.service
```

- Now

```
sudo mkinitcpio -P
```

- To rollback boot in one of the snapshot entries and do

```
sudo mount /dev/mapper/root /mnt
sudo mv /mnt/@ /mnt/@.broken
sudo btrfs subvolume snapshot /mnt/@snapshots/number/snapshot /mnt/@
sudo umount /mnt
```

- If you don't remember the number of the snapshot you want to rollback use

```
sudo grep -r '<date>' /mnt/@snapshots/*/info.xml
```

- Then reboot and after booting the new root partition if there is a problem with pacman use
`sudo rm -rf /var/lib/pacman/db.lck`